cellular thickening called filiform apparatus of synergids help in the absorption, conduction of nutrients from the nucellus to embryo sac. It also guides the pollen tube into the egg. Thus, a 7 celled embryo sac with 8 nuclei is formed. The structure of embryo sac is given in Figure 1.10.

1.4.3 Pollination



Pollination is a wonderful mechanism which provides food, shelter etc., for the pollinating animals. Many plants are pollinated by a particular animal species and the flowers are modified accordingly and thus there exists a co-evolution between plants and animals. Let us imagine if pollination fails. Do you think there will be any seed and fruit formation? If not what happens to pollinating organisms and those that depend on these pollinating organism for the food? Here lies the significance of the process of pollination.

The pollen grains produced in the anther will germinate only when they reach the stigma of the pistil. The reproductive organs, stamens and pistil of the flower are spatially separated, a mechanism which is essential for pollen grains to reach the stigma is needed. This process of transfer of pollen grains from the anther to a stigma of a flower is called **pollination.**

Pollination is a characteristic feature of spermatophyte (Gymnosperms and Angiosperms). Pollination in gymnosperms is said to be direct as the pollens are deposited directly on the exposed ovules, whereas in angiosperms it is said to be indirect, as the pollens are deposited on the stigma of the pistil. In majority of angiosperms, the flower opens and exposes its mature anthers and stigma for pollination. Such flowers are called **chasmogamous** and the phenomenon is **chasmogamy.** In other plants, pollination occurs without opening and exposing their sex organs. Such flowers are called **cleistogamous** and the phenomenon is **cleistogamy.**

Based upon the flower on which the pollen of a flower reaches, the pollination is classified into two kinds, namely, **self-pollination (Autogamy) and cross-pollination (Allogamy).**

A. Self-pollination or Autogamy (Greek Auto = self, gamos = marriage):

According to a majority of Botanists, the transfer of pollen on the stigma of the same flower is called **self-pollination or Autogamy**. Self-pollination is possible only in those plants which bear bisexual flowers. In order to promote self- pollination the flowers of the plants have several adaptations or mechanisms. They are:

1. Cleistogamy: In cleistogamy (Greek Kleisto = closed. Gamos = marriage) flowers never open and expose the reproductive organs and thus the pollination is carried out within the closed flower. *Commelina, Viola*, *Oxalis* are some examples for cleistogamous flowers. In *Commelina benghalensis,* two types of flowers are produced- aerial and underground flowers. The aerial flowers are brightly coloured, chasmogamous and insect pollinated. The underground flowers are borne on the subterranean branches of the rhizome that are dull, cleistogamous and self pollinated and are not dependent on pollinators for pollination.(Figure 1.11).

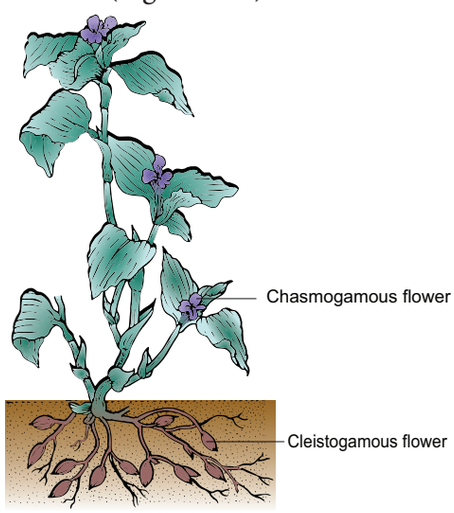


Figure 1.11 *Commelina* with Cleistogamous and Chasmogamous flowers

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